

Report: Neural Style Transfer Implementation and Findings

1. Introduction

Neural Style Transfer (NST) is a deep learning technique that enables the transformation of an image by applying the artistic style of another image while preserving the original content structure. In this project, we implemented NST using a pre-trained **VGG-19** model, optimizing a target image iteratively to minimize content and style losses.

2. Implementation Details

Our implementation involved several key components:

2.1 Feature Extraction using VGG-19

- We used the **VGG-19** network to extract feature representations from images.
- Specific layers were selected to compute **content loss** (conv4_2) and **style loss** (various convolutional layers such as conv1_1, conv2_1, etc.).

2.2 Content and Style Loss Computation

- **Content Loss:** Computed as the mean squared error (MSE) between the target and content feature maps.
- **Style Loss:** Computed using the Gram matrices of the feature maps.
 - The Gram matrix captures correlations between feature activations.
 - The loss is normalized by the number of feature channels.

2.3 Optimization Strategy

- The target image was optimized using the **Adam optimizer** with different learning rates.
- Experiments were conducted with different values of content weight (alpha) and style weight (beta).

3. Experimental Setup

To explore the effects of different hyperparameter settings, we conducted experiments with the following configurations:

Experiment	Learning Rate	Content Weight (\alpha)	Style Weight (\beta)	Steps
Exp 1	0.003	1e5	1e6	2000
Exp 2	0.001	1e4	1e5	1500
Exp 3	0.005	1e5	5e6	2500

4. Results and Analysis

Below are the key observations from the experiments:

4.1 Effect of Learning Rate

- Higher learning rates (e.g., **0.005**) led to faster convergence but sometimes caused instability.
- Lower learning rates (e.g., **0.001**) resulted in smoother transformations but required more iterations.

4.2 Effect of Content vs. Style Weight

- A **higher content weight (alpha)** preserved the original image structure better.
- A **higher style weight (beta)** introduced more artistic patterns but sometimes distorted content details.

4.3 Visual Comparison

The following images illustrate the style transfer results under different settings:

- **Figure 1:** Original Content Image
- **Figure 2:** Style Image
- **Figure 3:** Generated Image (Exp 1: Balanced content & style)
- **Figure 4:** Generated Image (Exp 2: More content retention)
- **Figure 5:** Generated Image (Exp 3: More stylistic influence)

5. Challenges and Future Work

5.1 Addressing Artifacts

- Some results showed artifacts, possibly due to overfitting on style loss.
- Applying **Total Variation (TV) Loss** can help smooth the results.

5.2 Alternative Architectures

- Future work could explore alternative architectures like **transformers for faster style transfer**.
- Investigating **adaptive instance normalization (AdaIN)** for real-time style transfer.

6. Conclusion

This project successfully implemented Neural Style Transfer, demonstrating how hyperparameters influence the quality of generated images. We found that adjusting **learning rates, content-to-style weight ratios, and iteration steps** significantly impacts the output quality. Future improvements could focus on refining stability and reducing computational costs.